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Abstract

In this study, biocarbon adsorbents were obtained from fennel and caraway seeds through microwave-assisted chemical activation with sodium carbonate. The activation process involved carbonizing the raw material at 300 °C for 30 min., followed by impregnation with sodium carbonate at a precursor-to-activator mass ratio of 1:2. Activation was performed at two distinct temperatures—500 °C and 600 °C—with an activation time of 15 min. The structural, textural, and surface chemical characteristics of the obtained biocarbons were investigated using complementary analytical techniques, including low-temperature nitrogen adsorption–desorption isotherms, X-ray photoelectron spectroscopy (XPS), scanning electron microscopy (SEM), X-ray diffraction (XRD), Boehm titration, and pH analysis of aqueous extracts. The resulting adsorbents demonstrated low development of specific surface area (109–154 m²/g) and limited sorption capacity for methylene blue (20–32 mg/g). Adsorption experiments indicated that the Freundlich isotherm model most accurately described the data, suggesting multilayer adsorption on heterogeneous surfaces. Thermodynamic evaluations showed the adsorption to be both spontaneous and endothermic. The adsorption mechanism is primarily governed by electrostatic interactions between the cationic dye and surface functional groups, π – π interactions with the carbon structure, and diffusion within mesopores. This study provides a comparative evaluation of microwave-assisted Na₂CO₃ activation of fennel and caraway seed waste and assesses the potential of these biochars for dye removal from aqueous solutions.

Keywords: biocarbon; caraway seed; fennel seed; chemical activation; methylene blue; thermodynamic study; kinetic study



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1. Introduction

The employment of microwave-assisted activation has emerged as a highly effective method for the production of activated carbon and biochar adsorbents, particularly for applications requiring rapid heating and reduced energy consumption. This approach employs microwave radiation to expeditiously and uniformly heat carbonaceous precursors, resulting in the formation of porous structures suitable for adsorption processes [1]. Unlike conventional thermal methods, microwave heating enables volumetric and selective energy transfer, which promotes faster carbonization and more homogeneous pore development [2]. Consequently, microwave-assisted activation has attracted increasing attention as a sustainable alternative for producing functional adsorbents with reduced

processing time and lower energy demand [3,4]. Activated carbon is widely recognized for its high surface area and porous structure, making it an excellent material for adsorption in diverse fields such as water treatment, gas storage, and contaminant removal [1–5]. Generally, activated carbon is produced through two main approaches: physical activation and chemical activation. The latter involves the impregnation of a carbon-rich precursor with a chemical agent, followed by thermal treatment to promote pore formation [5]. Common chemical activators such as KOH, ZnCl₂, and H₃PO₄ are highly effective in generating microporous structures but suffer from disadvantages including high corrosiveness, environmental burden, and the need for extensive post-treatment washing. In contrast, sodium carbonate (Na₂CO₃) is a milder and more environmentally benign activating agent, characterized by lower toxicity and safer handling [6,7]. However, due to its lower chemical reactivity, Na₂CO₃ typically exhibits limited ability to etch the carbon matrix, which may result in reduced surface area and a predominance of mesoporous structures compared to strongly activating agents. From a kinetic and thermodynamic perspective, Na₂CO₃-activated carbons are therefore expected to favor physisorption mechanisms and diffusion-controlled mass transfer, rather than strong surface–adsorbate interactions.

The thermodynamic behavior of adsorption provides insight into the energetic feasibility and dominant mechanisms governing pollutant uptake by Na₂CO₃-activated carbons. In aqueous systems, negative Gibbs free energy (ΔG) values indicate spontaneous adsorption, while relatively low enthalpy changes (ΔH) are generally associated with physical adsorption driven by van der Waals forces and electrostatic interactions. In such mildly activated carbons, surface functional groups and pore accessibility play a more significant role than total surface area alone [8,9]. Similarly, adsorption kinetics in Na₂CO₃-derived carbons is often controlled by intraparticle diffusion and external mass transfer limitations, due to the prevalence of mesopores and moderate microporosity. Understanding these kinetic and thermodynamic features is essential for evaluating the real applicability of Na₂CO₃-activated biochars in water treatment processes.

The growing demand for sustainable environmental technologies has intensified interest in developing low-cost adsorbents derived from renewable resources. Agricultural and herbal wastes represent valuable carbon-rich precursors for producing activated carbons, contributing simultaneously to waste valorization and circular economy strategies [10–12]. Despite extensive research on various biomass sources, studies focusing on fennel (*Foeniculum vulgare*) and caraway (*Carum carvi*) seeds as precursors for activated carbon remain scarce. These materials are rich in lignocellulosic components and essential oil residues, which may influence the surface chemistry and adsorption behavior of the resulting biochars. Previous studies have demonstrated that fennel and caraway seeds are promising precursors for the production of activated carbons, particularly when strong chemical activators such as potassium carbonate or phosphoric acid are employed. For instance, chemically activated fennel-based carbons obtained using K₂CO₃ exhibited very high specific surface areas (up to 1052 m²/g) and excellent adsorption capacities toward methylene blue (up to 454 mg/g) and iodine (up to 1215 mg/g) [13]. Similarly, caraway-derived carbons activated with K₂CO₃ and H₃PO₄ showed surface areas up to 926 m²/g and methylene blue adsorption capacities reaching 296 mg/g [14]. Other studies have also confirmed that both fennel- and caraway-based carbons can achieve high sorption performance when strong chemical activation and conventional heating methods are applied, with methylene blue capacities typically ranging from 204 to 328 mg/g [15]. In contrast, studies employing sodium carbonate as the activating agent consistently report lower surface areas and adsorption capacities, reflecting the milder activating nature of Na₂CO₃. For example, fennel-derived carbons activated with Na₂CO₃ under conventional conditions exhibited significantly lower iodine and methylene blue uptake compared to

K₂CO₃-activated counterparts [13,15]. Microwave-assisted Na₂CO₃ activation of fennel and caraway seeds resulted in materials with very low surface areas (25–26 m²/g) and moderate adsorption capacities (68–91 mg/g for methylene blue) [16].

Methylene blue (MB) was selected as the model pollutant in this study due to its widespread use in the textile, pharmaceutical, and paper industries. MB is a chemically stable cationic dye that is frequently detected in industrial effluents and exhibits resistance to conventional biological degradation. Exposure to methylene blue may cause adverse health effects, including skin irritation, respiratory problems, and gastrointestinal disorders. For these reasons, MB is commonly employed as a benchmark compound in adsorption studies, enabling reliable comparison of adsorbent performance across different materials [13,17].

The combined use of fennel and caraway biomass, sodium carbonate activation, and microwave treatment represents a sustainable and energy-efficient strategy for producing functional biochar adsorbents. The selection of Na₂CO₃ is motivated by its environmental compatibility and reduced chemical hazards, while microwave heating enables rapid and uniform activation at relatively low temperatures. Fennel and caraway seeds are abundant herbal industry by-products, and their intrinsic chemical composition may promote the formation of oxygen-containing functional groups, potentially compensating for the relatively low specific surface area through enhanced surface chemistry and adsorption affinity. The novelty of the present study lies in the application of microwave-assisted Na₂CO₃ activation to fennel and caraway seeds for the synthesis of biochar adsorbents and the systematic evaluation of their textural, chemical, and adsorption properties. Compared to previously reported Na₂CO₃-based systems [16], the materials obtained in this work demonstrate improved textural properties. This highlights the potential of microwave-assisted carbonate activation as an energy-efficient and environmentally benign route for producing functional adsorbents.

2. Results and Discussion

2.1. Physicochemical Characterization of the Activated Carbons

Figure 1 presents the nitrogen adsorption–desorption isotherms (A) and corresponding pore size distributions (B) of the samples, recorded at −196.15 °C. According to the IUPAC classification, adsorption isotherms exhibiting Type IV curves with H2-type hysteresis loops are characteristic of mesoporous materials [18]. The isotherms obtained show this behavior, indicating the development of mesopores with constricted necks and wider pore bodies—often referred to as ink-bottle-type pores [18]. The observed capillary condensation in this pore size range further supports the presence of well-defined mesoporosity.

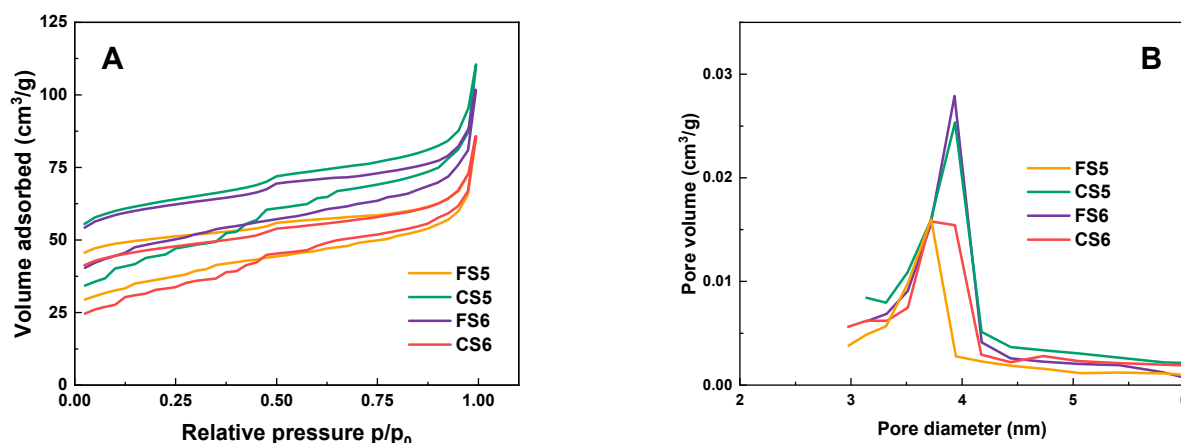


Figure 1. Low-temperature N₂ adsorption–desorption isotherms (A) and pore size distribution (B) of biocarbon samples obtained.

Table 1 shows the textural parameters, ash content and iodine number of the activated carbons obtained. The analysis of the data presented in Table 1 shows that the specific surface area of the carbon materials obtained ranges from 109 to 154 m²/g, with the pore size ranging from 4.09 to 4.87 nm. In the case of fennel seeds, an increase in activation temperature by 100 °C resulted in a 37 m²/g improvement in specific surface area and an enlargement of the surface area of the micropores. Although micropores are present in all samples, as confirmed by microporous surface areas ranging from 47 to 75 m²/g and micropore volumes of 0.020–0.038 cm³/g, their contribution to total porosity remains moderate ($V_m/V_t = 0.150$ – 0.241). The calculated average pore diameters (4.09–4.87 nm) indicate that mesopores constitute the dominant pore fraction. This pore structure is consistent with the mild activation mechanism of Na₂CO₃, which promotes limited carbon matrix etching and preferential pore widening rather than extensive micropore generation. Considering that the molecular size of methylene blue is approximately 1–1.5 nm, the presence of mesopores may facilitate diffusion of dye molecules into the pore network, while adsorption is likely governed predominantly by surface functional groups rather than by micropore filling.

Table 1. Textural parameters, iodine number and ash content of obtained biocarbon samples.

Sample	Surface Area ¹ (m ² /g)		Pore Volume (cm ³ /g)		V_m/V_t	Average Pore Size (nm)	Iodine Number (mg/g)	Ash Content (%)
	Total	Microporous	Total	Microporous				
FS5	117	52	0.133	0.027	0.239	4.53	135	7.86
CS5	146	75	0.171	0.032	0.187	4.68	116	8.09
FS6	154	68	0.158	0.038	0.241	4.09	141	8.97
CS6	109	47	0.133	0.020	0.150	4.87	101	9.21

¹ Error range between 2 and 5%.

Furthermore, an increase in iodine number from 135 to 141 mg/g was observed. Conversely, in the case of caraway seeds, a contradictory relationship was observed. Increasing the temperature resulted not only in a decrease in the total adsorbent area, but also in the micropore area and a decrease in the iodine number. The adsorption of iodine on carbon adsorbents enables the assessment of their capacity to eliminate pollutants with molecular diameters approaching 1 nm [19,20]. The iodine number is a simple, standardized, and reliable indicator of micropore content and adsorption capacity, which are key properties governing the performance of activated carbons [21]. The slight increase in ash content observed with rising activation temperature (approximately 1% per 100 °C) may be attributed to the enhanced thermal degradation of the organic matrix. At higher temperatures, more volatile organic components are removed, concentrating the thermally stable mineral matter in the final product. While this trend is relatively modest, elevated ash content can negatively affect adsorbent performance by occupying active sites or blocking pores, potentially reducing overall surface area and adsorption efficiency [22]. Therefore, controlling ash content is essential to maintaining optimal material quality, particularly when designing adsorbents for applications involving low-concentration contaminants in aqueous systems. It is evident from the average pore diameter values that all materials exhibit a predominantly mesoporous structure. The results obtained demonstrate that sodium carbonate was ineffective in significantly developing the specific surface area of the carbonaceous materials. Furthermore, an increase in activation temperature did not result in a substantial enhancement of the surface area. This behavior can be attributed to the relatively low chemical reactivity of Na₂CO₃ compared to conventional activating agents such as KOH, ZnCl₂, or H₃PO₄ [6].

The obtained biocarbon samples can be compared to adsorbents described in the literature. An adsorbent derived from *Astragalus membranaceus* through activation with Na_2CO_3 at 800°C exhibited a total surface area of $267\text{ m}^2/\text{g}$ [23]. Sample obtained from caraway seeds through activation with Na_2CO_3 (precursor:activator ratio: 1:2) at 700°C in conventional furnace exhibited similar total surface area of $269\text{ m}^2/\text{g}$ [14]. However, in spite of the fact that biocarbon samples obtained from fennel and caraway seeds in microwave-assisted chemical activation exhibit a total surface area lower by about $100\text{ m}^2/\text{g}$, they have been obtained at a temperature lower, by about $300\text{--}400^\circ\text{C}$.

X-ray photoelectron spectroscopy (XPS) was utilized to investigate the surface elemental composition in detail. High-resolution spectra for the C1s and O1s regions are shown in Figure 2.

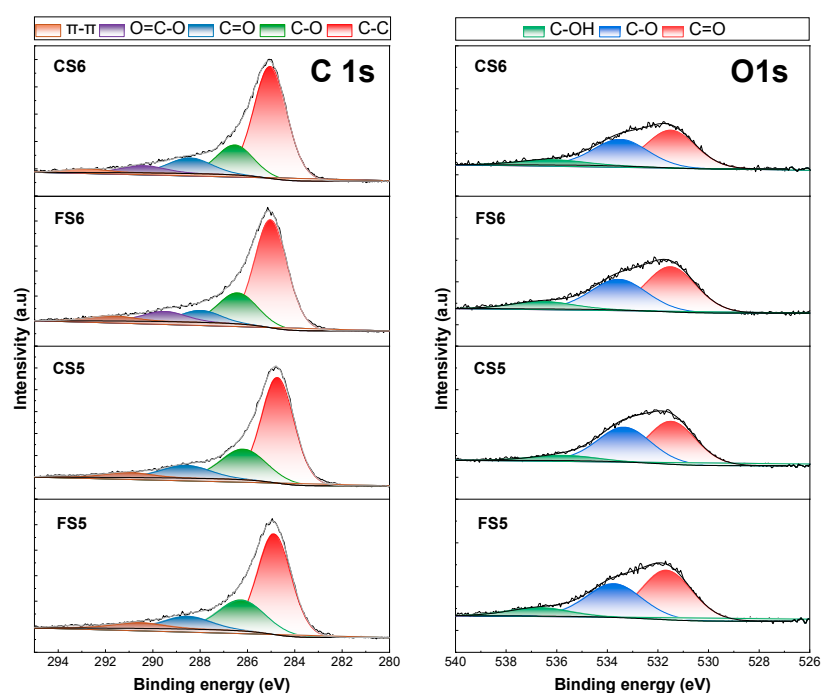


Figure 2. The XPS carbon (C1s) and oxygen (O1s) spectra of obtained biocarbon samples.

The C1s spectrum was deconvoluted into four to five distinct peaks centered around 284.8, 286.1, 287.6, 289.5, and 291.4 eV, which correspond to aromatic C-C bonds, aromatic C-O groups, ketone C=O functionalities, carboxylic O=C-O groups, and π - π interactions, respectively [24]. The C1s spectra of the samples are largely comparable; however, a notable difference is observed in the spectra of the samples obtained at higher temperatures (CS6 and FS6). These samples exhibit the presence of carboxylic O=C-O groups on their surface, a grouping that is absent in the C1s spectra of the other two samples. The O1s spectra were deconvoluted into three distinct peaks. The peaks in the range of 531.0–532.0 eV are attributed to oxygen double-bonded to carbon in carbonyl groups (C=O), those at 532.9–533.4 eV correspond to C-O groups, and the peaks between 534.8 and 536.7 eV are associated with hydroxyl groups (C-OH) [24]. The surface atomic O/C ratios followed the order FS5 (0.15) > FS6 (0.13) > CS6 (0.11) > CS5 (0.10), suggesting that fennel seed-derived carbons possess a relatively higher concentration of oxygen-containing surface groups compared to caraway seed-derived samples.

The acid–base properties of activated carbon surfaces play a crucial role in the adsorption of contaminants from water. These surface characteristics, which arise from the material's chemical composition, influence the interactions between the adsorbent and adsorbate, ultimately affecting the efficiency and mechanisms of pollutant removal [25].

Figure 3 illustrates the acid–base behavior of the biocarbon samples produced in this study. These properties were evaluated by measuring the pH of their aqueous extracts and quantifying the surface oxygen-containing functional groups through the Boehm titration method. The study demonstrated that sorptive materials which were chemically activated with sodium carbonate at a temperature of 500 °C exhibited a predominance of acidic groups; conversely, samples which were activated at 600 °C were characterised by a predominance of basic functional groups. This shift in surface chemistry can be attributed to thermal decomposition and rearrangement of functional groups during activation. At lower activation temperatures, oxygen-containing acidic groups such as carboxylic and phenolic functionalities are more stable and remain on the surface. However, as the temperature increases to 600 °C, these labile acidic groups undergo decomposition and volatilization, reducing their abundance [26]. Concurrently, higher temperatures favor the formation of more thermally stable basic groups, such as pyrone, chromene-type structures, and surface-bound alkali metal oxides introduced by the activating agent [27]. This transformation results in a relative increase in basic surface character, which can influence both the adsorption mechanism and the affinity toward specific contaminants. This is correlated with the pH of their aqueous extracts, which decreases in the order FS5 > CS5 > FS6 > CS6.

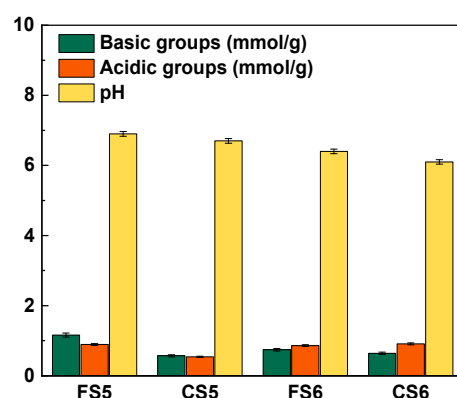


Figure 3. Acid–base properties of the obtained biocarbon samples.

The pH_{pzc} values of the samples were determined using the drift method, as shown in Figure 4. The pH_{pzc} values were 7.79 for FS5, 7.68 for CS5, 7.33 for FS6 and 6.98 for CS6. It can be concluded that the pH_{pzc} value of the resulting adsorbent increased with increasing activation temperature. As can be seen in Figure 4, the pH_{pzc} values correspond to the acid–base properties. According to the definition of the point of zero charge, activated carbon surfaces are negatively charged when $\text{pH} > \text{pH}_{\text{pzc}}$ and positively charged below this value.

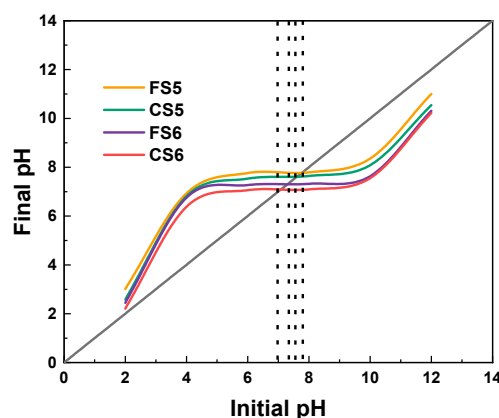


Figure 4. The point of zero charge (pH_{pzc}) of the obtained biocarbon samples.

A scanning electron microscope (SEM) was employed to conduct a comprehensive analysis of the surface morphology of the adsorbents. As demonstrated in Figure 5, the presence of brighter fragments in activated carbons may be attributed to the presence of ash.

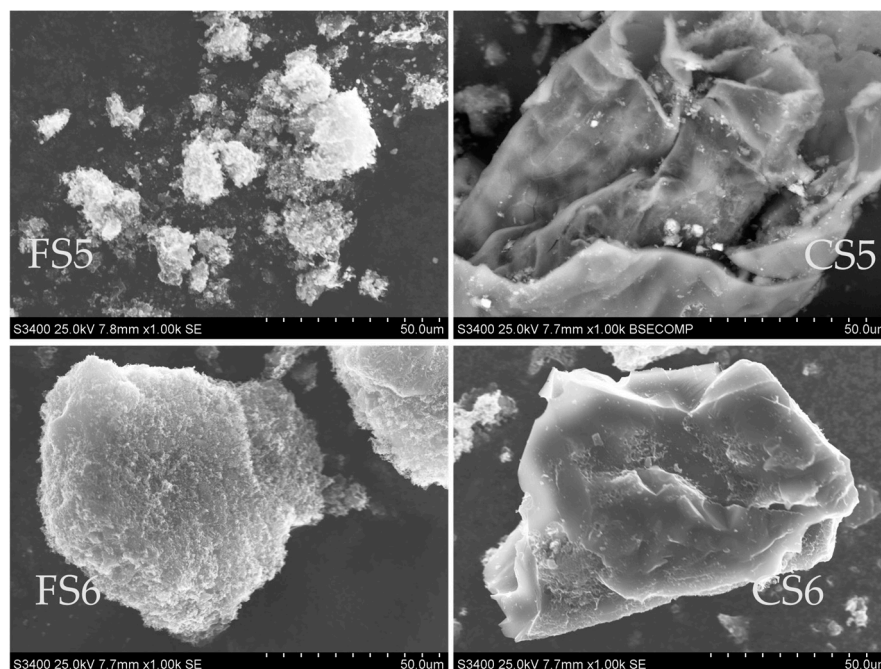


Figure 5. SEM images.

The carbonaceous sorbents obtained by chemical activation of the chosen biomass were subjected to X-ray diffraction (XRD) analysis, the results of which are presented in Figure 6. XRD patterns of the activated carbons show two broad reflections at $\sim 24^\circ$ and $\sim 43^\circ$ (2θ), indexed to the (002) and (100) planes of turbostratic carbon, indicating predominantly amorphous structure with small graphitic microdomains [28,29]. FS5 and FS6 exhibit nearly identical peak positions and widths, suggesting comparable interlayer spacing and crystallite dimensions.

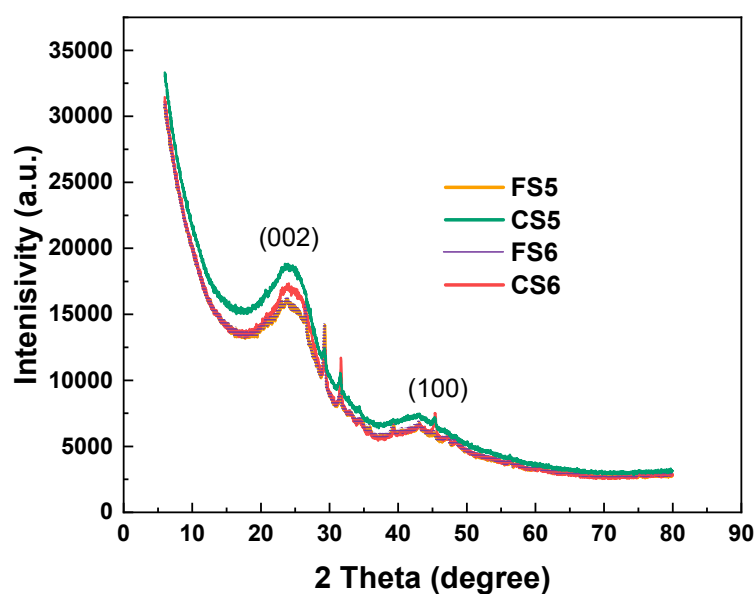


Figure 6. XRD patterns of biocarbons.

2.2. Adsorption

Figure 7 shows the adsorption isotherms of the obtained samples for methylene blue. The presented data indicates that the FS6 sample demonstrated the best sorption abilities for the tested pollutant. Furthermore, it was observed that biocarbon samples obtained from fennel seeds exhibited superior sorption properties in comparison to adsorbents derived from caraway seeds. The maximum sorption capacities of the samples FS5, CS5, FS6, and CS6 are 28, 24, 32 and 20 mg/g, respectively. It is noteworthy that the sorption capacities of the adsorbents exhibited a direct correlation with their iodine numbers. It is evident that an increase in temperature of 100 °C results in an adsorbent capacity of 4 mg/g higher for fennel seeds. Conversely, an opposing observation is seen for caraway seeds.

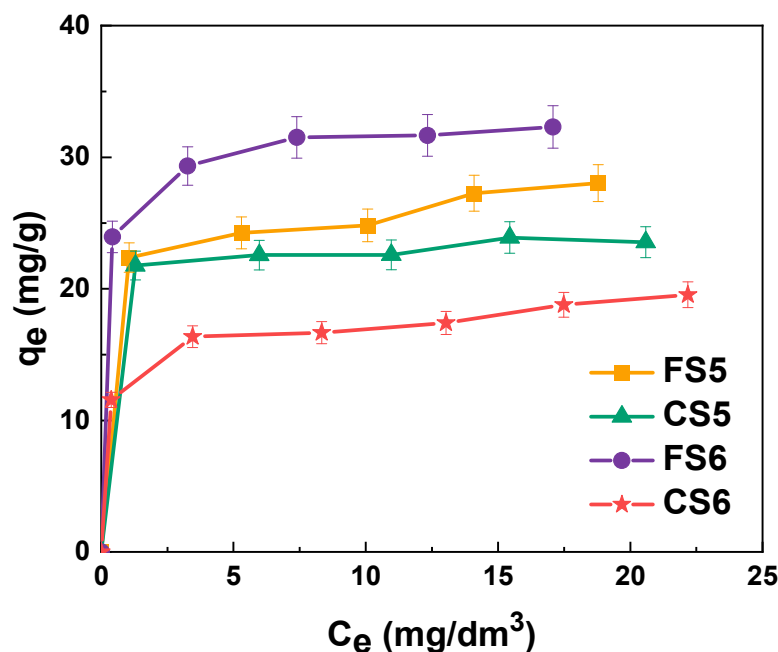


Figure 7. Isotherms of methylene blue adsorption on obtained biocarbon samples (adsorbent mass: 20 mg, volume of dye solution: 0.05 dm³).

The maximum adsorption capacities obtained in this study (20–32 mg/g) are lower than those reported for highly developed KOH- or K₂CO₃-activated carbons, which often exceed 200–500 mg/g [30]. This difference can be attributed to the relatively modest specific surface areas of the Na₂CO₃-activated biochars (109–154 m²/g). Sodium carbonate is a milder activating agent that promotes limited carbon matrix etching and predominantly mesoporous structures under microwave conditions. Consequently, adsorption in the present materials is governed primarily by surface chemistry and electrostatic interactions rather than extensive microporosity development. Although these materials are not competitive with high-temperature KOH-activated carbons in terms of maximum capacity, they demonstrate functional adsorption performance achieved at significantly lower activation temperatures (500–600 °C) and shorter processing times. This highlights the trade-off between adsorption capacity and energy demand, emphasizing the potential of microwave-assisted Na₂CO₃ activation as a more energy-efficient and environmentally benign approach. To better position the obtained materials within the current research landscape, a comparative analysis of methylene blue adsorption capacities reported for biomass-derived activated carbons is presented in Table 2.

Table 2. Comparison of the various reported adsorbents and their sorption capacities for methylene blue.

Precursor	Activator	Precursor: Activator Ratio	Activation Time (min)	Activation Temperature (°C)	Sorption Capacity (mg/g)	Source
fennel seeds	Na ₂ CO ₃	1:2	45	700	77	[13]
caraway seeds	Na ₂ CO ₃	1:2	45	700	37	[14]
fennel seeds	Na ₂ CO ₃	1:1	45	700	11	[15]
raspberry fruit	Na ₂ CO ₃	1:2	60	700	146	[31]
blackcurrant fruit	Na ₂ CO ₃	1:2	60	700	137	
nettle leaves	Na ₂ CO ₃	1:2	60	700	96	
green tea leaves	Na ₂ CO ₃	1:2	60	700	85	[32]
pineapple peel	KOH	-	120	700	136	
<i>Saponaria officinalis</i> root	H ₂ SO ₄	1:1	90	650	75	

The comparison presented in Table 2 demonstrates that the adsorption performance of carbon materials toward methylene blue strongly depends on precursor type, activating agent, activation temperature, and processing time. In the present study, the maximum sorption capacities were between 20 and 32 mg/g. When compared with literature data, the adsorption capacities obtained in this study are lower than those reported for Na₂CO₃-activated carbons prepared at 700 °C for longer activation times (45–60 min), where sorption capacities ranged from 77 to 146 mg/g depending on the biomass source. The higher performance of these materials can be attributed to more severe activation conditions, which promote greater micropore development and surface area expansion. Similarly, carbons activated with stronger agents such as KOH achieved capacities of up to 136 mg/g, reflecting the higher efficiency of aggressive chemical activation in generating well-developed porous structures. However, it is important to emphasize that the materials investigated in the present work were synthesized under milder microwave-assisted conditions (500–600 °C, 15 min), which significantly reduce processing time and energy input. The moderate adsorption capacities observed (20–32 mg/g) are consistent with the limited micropore development and predominance of mesoporosity characteristic of Na₂CO₃ activation under reduced thermal severity. Therefore, while the sorption performance is lower than that of highly activated carbons produced at 700 °C, the present approach offers a more energy-efficient and environmentally benign alternative for producing functional adsorbents.

Using the experimental results, parameters for two linear adsorption models, the Langmuir and Freundlich isotherms, were derived, as illustrated in Figure 8. Table 3 summarizes the calculated constants associated with both models. The Langmuir model assumes monolayer adsorption on a homogeneous surface with a finite number of adsorption sites [34].

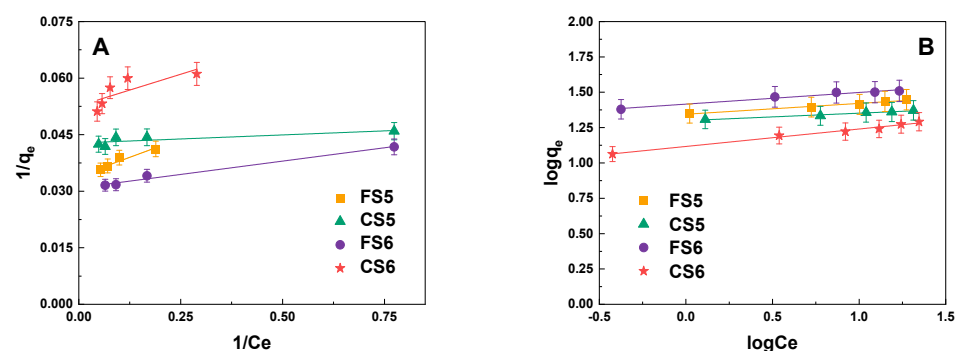
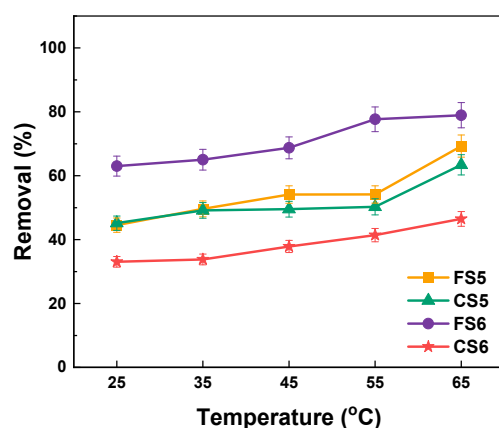
**Figure 8.** Linear fitting for methylene blue on obtained biochar to (A) Langmuir model and (B) Freundlich model.

Table 3. The values of constants determined for the linear Langmuir and Freundlich models for experimental data of methylene blue.

Isotherm	Parameters	Sample			
		FS5	CS5	FS6	CS6
Langmuir	q_{exp} (mg/g)	28	24	32	20
	K_L (dm ³ /mg)	0.856	9.974	2.184	1.569
	q_m (mg/g)	29	23	32	19
	R^2	0.934	0.656	0.986	0.615
	Adj^2	0.901	0.541	0.979	0.487
Freundlich	K_F (mg/g (dm ³ /mg) ^{1/n})	22.107	19.923	26.065	13.081
	1/n	0.077	0.053	0.082	0.123
	R^2	0.979	0.972	0.971	0.986
	Adj^2	0.973	0.962	0.962	0.983

The adsorption data were analyzed using both Langmuir and Freundlich isotherm models (Figure 8, Table 3). Although the Langmuir model provided a reasonable fit for FS5 ($R^2 = 0.934$) and FS6 ($R^2 = 0.986$), lower correlation coefficients were obtained for CS5 ($R^2 = 0.656$) and CS6 ($R^2 = 0.615$), indicating that the assumption of homogeneous monolayer adsorption is not fully valid for all samples. In contrast, the Freundlich model exhibited consistently high correlation coefficients for all adsorbents ($R^2 = 0.971$ – 0.991), demonstrating superior agreement with experimental data. This suggests that methylene blue adsorption occurs predominantly on heterogeneous surfaces through multilayer physisorption. The Freundlich model describes adsorption on a heterogeneous surface with varying binding energies and the possibility of multilayer adsorption [35]. The heterogeneity arises from the distribution of pore sizes and the presence of diverse oxygen-containing functional groups, as confirmed by XPS and Boehm titration analyses. The Freundlich parameter $1/n$ (<1 for all samples) further confirms favorable adsorption conditions and surface heterogeneity.

The influence of temperature on the methylene blue sorption capacity of the produced biochars was examined. The results shown in Figure 9 suggest that temperature had a minimal effect on the adsorption performance of the materials tested. It is evident from the data that there is a direct correlation between temperature and the rate of removal, with the largest increase in removal percentage observed for sample FS6 when the temperature is increased by 10 °C, between 45 °C and 55 °C (9.0 percentage points).

**Figure 9.** Effect of temperature of the aqueous solution of dye on removal (adsorbent mass: 20 mg, volume of dye solution: 0.05 dm³, pH of the solution: 6.3, dye concentration: 20 mg/dm³).

For the other samples, the largest change in removal percentage is observed when the temperature is increased from 55 °C to 65 °C (5.0–15.1 percentage points). The elevation in temperature from 25 to 65 °C resulted in a substantial enhancement in methylene blue removal (%) for sample FS5 (24.7 percentage points), while the least significant increase was observed for sample CS6 (13.4 percentage points). This minor impact on the removal of contaminants from the aqueous phase is economically advantageous, as the materials can reach their sorption capacities at ambient temperature. The thermodynamic parameters calculated from the experimental data are summarized in Table 4.

Table 4. Thermodynamic parameters of adsorption on the obtained biochar.

Sample	Temperature (°C)	ΔG° (kJ/mol)	ΔH° (kJ/mol)	ΔS° (J/mol $\times$ K)
FS5	25	−1.79	21.65	77.92
	35	−2.24		
	45	−2.86		
	55	−2.95		
	65	−4.86		
CS5	25	−1.79	15.86	59.07
	35	−3.76		
	45	−2.45		
	55	−2.45		
	65	−2.48		
FS6	25	−3.59	16.53	67.52
	35	−3.94		
	45	−4.51		
	55	−5.90		
	65	−6.29		
CS6	25	−0.52	15.08	50.91
	35	−0.63		
	45	−1.11		
	55	−1.55		
	65	−2.18		

Analysis of these results indicates that the adsorption of the dye onto all the synthesized adsorbents was spontaneous, as demonstrated by the negative ΔG° values [36]. The positive ΔH° values suggest that the adsorption process is endothermic, involving the uptake of heat during interaction between the dye molecules and the adsorbent. Additionally, the relatively low adsorption energies observed for the samples point to a predominantly physical adsorption mechanism. Physisorption involves weak van der Waals forces between the adsorbate and the adsorbent surface, distinguishing it from chemisorption, which is characterized by stronger chemical bond formation. This difference is reflected in the lower enthalpy changes and the reversible nature of physisorption [37,38]. Based on the data shown in Table 4, it can be concluded that physisorption occurred on all bio-carbon samples. One of the primary mechanisms of physisorption is London dispersion forces, which arise from temporary dipoles induced by fluctuations in electron clouds [39]. Another important mechanism is dipole-induced dipole interactions, where a polar adsorbate induces a dipole moment in a nonpolar surface, enhancing attraction. In some cases, electrostatic interactions contribute to physisorption, especially when the adsorbate carries a permanent dipole moment and interacts with charged or polarizable surfaces [39]. Physisorption generally occurs in multiple layers (multilayer adsorption), as described by the BET (Brunauer–Emmett–Teller) isotherm, which extends the Langmuir model to account for successive layers of adsorption [40]. The weak nature of physisorption allows

easy desorption upon changes in temperature or pressure, making it crucial in applications like gas storage, heterogeneous catalysis, and adsorption-based separation processes [41].

Comparing the thermodynamic data presented in Table 4 with previously published results, it can be concluded that samples obtained from caraway seeds through activation with Na_2CO_3 (precursor:activator ratio: 1:2) at 700 °C in a conventional furnace are characterised by much lower ΔG° values (-7.34 – -37.44) and much higher values of ΔH° (217.16) and ΔS° (753.96) [14]. The higher activation temperature (700 °C vs. 500–600 °C) and conventional furnace method likely generate biochars with significantly different surface chemistry and porosity. Higher temperatures typically create more developed pore structures and a higher degree of carbonization, which may increase adsorption capacity but also lead to stronger adsorbate-adsorbent interactions reflected in higher ΔH° . Higher ΔS° values in the referenced study suggest a greater increase in randomness or disorder at the solid–liquid interface during adsorption, possibly due to more extensive structural rearrangements or release of solvent molecules upon dye binding.

The influence of contact time between the adsorbent and methylene blue on the sorption capacity of the activated carbons was examined (Figure 10). The data show that adsorption equilibrium for the chemically activated biocarbon is achieved after about 3 h, which is advantageous from a cost-efficiency standpoint. These observations were used to determine parameters for four kinetic models: pseudo-first-order, pseudo-second-order, intraparticle diffusion, and Elovich, as summarized in Table 5. The resulting graphs for linear fitting of these models are shown in Figure 11. The correlation coefficient R^2 for the pseudo-first-order model ranges from 0.890 to 0.963, for intraparticle diffusion from 0.935 to 0.965, and for Elovich from 0.976 to 0.998. For the pseudo-second-order model, the value is not less than 0.997. This suggests that the adsorption of dye molecules on the obtained adsorbents follows the pseudo-second-order model. This conclusion is further supported by the theoretical sorption capacity ($q_{e/\text{cal}}$), which aligns with the experimental values. The higher the value of pseudo-second-order kinetics constant, the faster the reaction between adsorbent and adsorbate [42].

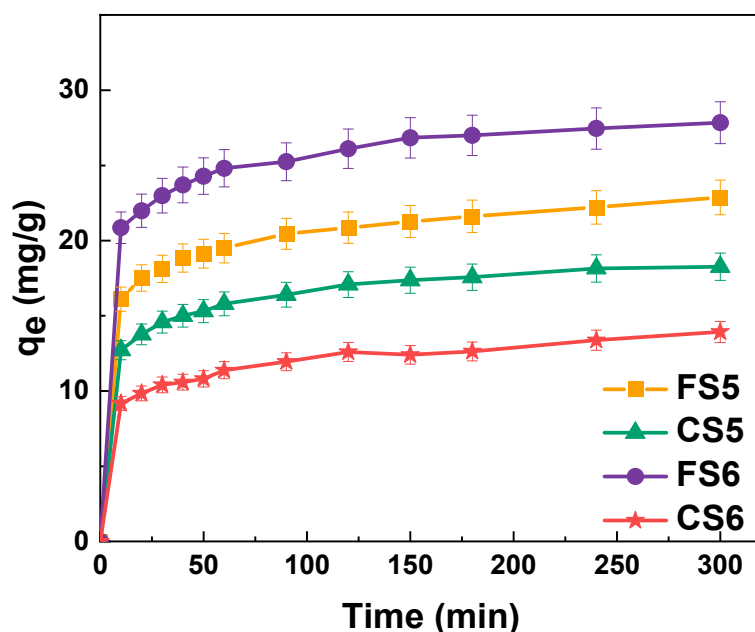
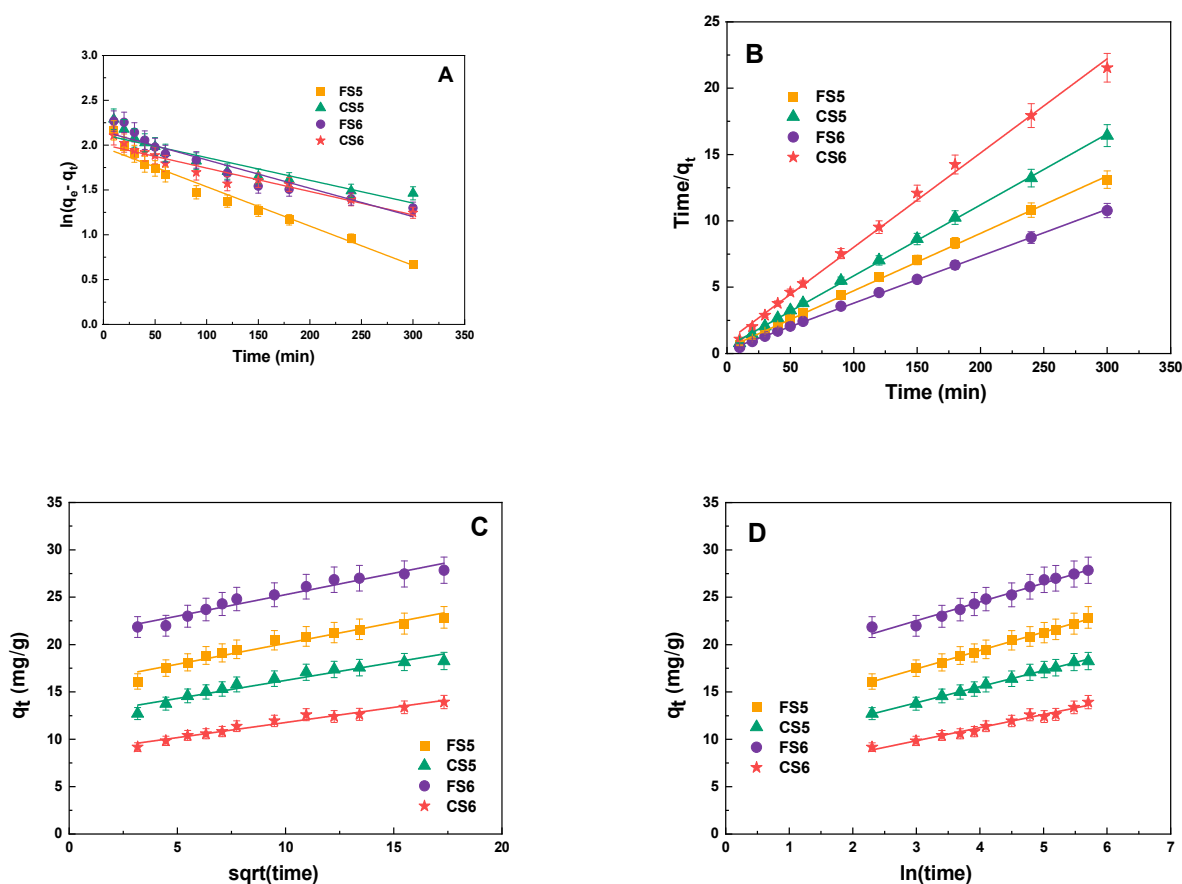


Figure 10. Effect of contact time on the sorption capacities of the biocarbon samples (adsorbent mass: 20 mg, pH of the solution: 6.3, volume of dye solution: 0.05 dm³, dye concentration: 20 mg/dm³).

Table 5. Kinetic model parameters.

Kinetics Model	Parameters	Sample			
		FS5	CS5	FS6	CS6
Pseudo-first-order	q_e (mg/g)	25	23	32	17
	k_1 (1/min)	1.563×10^{-5}	9.133×10^{-6}	9.528×10^{-6}	7.583×10^{-6}
	R^2	0.963	0.890	0.922	0.940
	$q_{e/cal}$ (mg/g)	7	9	9	8
Pseudo-second-order	k_2 (g/mg $\times$ min)	4.484×10^{-3}	4.478×10^{-3}	5.739×10^{-3}	3.307×10^{-3}
	R^2	0.999	0.999	0.999	0.997
	$q_{e/cal}$ (mg/g)	23	19	28	14
Intraparticle diffusion	k_{id} (mg/g $\times$ min $^{1/2}$)	0.440	0.382	0.453	0.319
	R^2	0.952	0.935	0.950	0.965
	C (mg/g)	15.723	12.399	20.739	8.581
Elovich	α (mg/g $\times$ min)	766.098	295.680	8694.073	84.810
	R^2	0.998	0.997	0.976	0.977
	β (g/mg)	0.515	0.584	0.506	0.722

**Figure 11.** Linear fitting for methylene blue on obtained biochar to (A) pseudo-first-order model, (B) pseudo-second-order model, (C) intraparticle diffusion and (D) Elovich model.

The results in Table 5 indicate that the adsorption rate increased in the series: CS6 < CS5 < FS5 < FS6. The Elovich isotherm model has been extensively employed to facilitate the understanding and analysis of the multilayer adsorption process of substances upon heterogeneous surfaces [43,44]. The model is predicated on the kinetic principle,

which posits the hypothesis that adsorption sites increase exponentially with adsorption [45]. The Elovich initial sorption rate constant (α) is highest for the FS6 sample and lowest for the CS6 sample. Notably, this value correlates with the adsorption rate constant (k_2) and the sorption capacities of the tested samples. Moreover, an inverse relationship is observed for the desorption β constant. In addition to the excellent fit of the pseudo-second-order model ($R^2 = 0.997\text{--}0.999$), the intraparticle diffusion model was applied to further elucidate the adsorption mechanism. The intercept values ($C = 8.581\text{--}20.739$ mg/g) are different from zero, confirming that intraparticle diffusion is not the sole rate-limiting step. Instead, the adsorption process is influenced by both boundary layer (external mass transfer) effects and diffusion within the pore structure. The magnitude of C varies among the samples, with FS6 showing the highest value, suggesting a more pronounced contribution of surface adsorption or boundary layer resistance in this case. These results indicate that methylene blue adsorption onto the investigated biocarbons proceeds through a combined mechanism involving surface adsorption followed by gradual diffusion into the internal pore network.

The effect of solution pH on the removal efficiency of methylene blue by the resulting biocarbon samples was investigated. The results are exhibited in Figure 12.

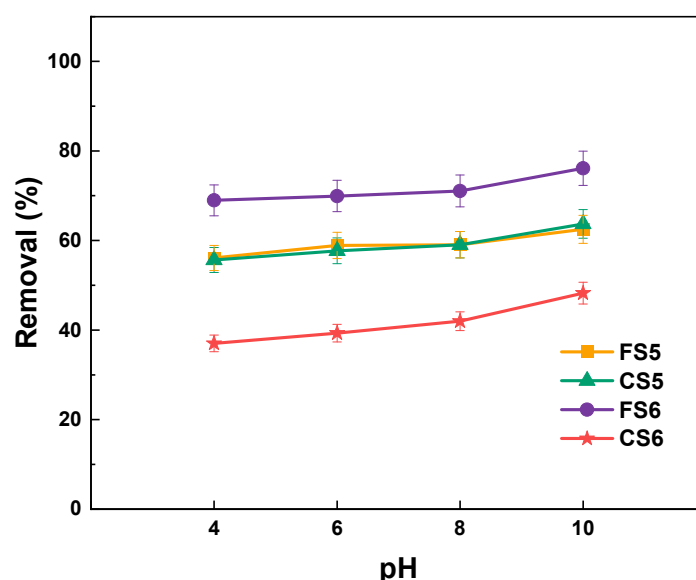


Figure 12. Effect of pH of the aqueous solution of dye on removal (adsorbent mass: 20 mg, volume of dye solution: 0.05 dm³, dye concentration: 20 mg/dm³).

The findings suggest that, for all samples, an increase in pH is associated with an increase in sorption capacity. A slight decrease in sorption capacity was observed for all adsorbents at pH 4 compared to other pH values. At higher pH levels, the adsorbents acquire a negative surface charge, enhancing the adsorption of positively charged dye molecules such as methylene blue [46]. Furthermore, research indicates that the optimal pH range for methylene blue adsorption is between 6 and 11, which aligns with the results obtained for biochar derived from fennel and caraway seeds [47].

An investigation was conducted into the effect of ionic strength on the efficiency of methylene blue removal. The effect of inorganic salt (NaCl) on the adsorption rate of MB on obtained biochar is demonstrated in Figure 13.

The increase in methylene blue removal with increasing ionic strength for FS6 and especially CS6 suggests that electrostatic repulsion between adsorbed cationic dye molecules plays a significant role. The addition of NaCl screens these repulsive interactions, allowing higher surface packing density. In contrast, the negligible effect observed for FS5 and

CS5 indicates that non-electrostatic interactions dominate the adsorption mechanism in these materials [48].

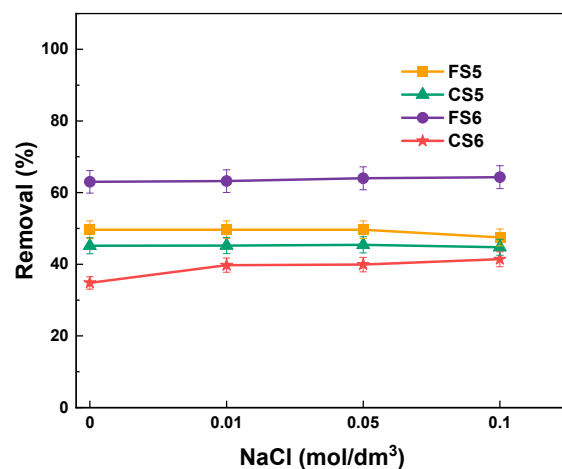


Figure 13. Effect of ionic strength (adsorbent mass: 20 mg, pH of the solution: 6.3, volume of dye solution: 0.05 dm³, dye concentration: 20 mg/dm³).

2.3. Desorption

Desorption studies were conducted using three eluents: 0.1 M HCl/NaOH and ethanol. Figure 14 shows bar charts of the % removal of dye after each cycle, as well as the adsorption capacity of the carbon samples (mg/g), represented by dots. The adsorbents were subjected to a regeneration process, following the removal of methylene blue by adsorption. Thereafter, the adsorbents were reused for further adsorption cycles. Figure 15 shows O1s spectra of FS6 sample after adsorption (D) and desorption (A–C) of methylene blue. Table 6 exhibits relative contents of oxygen and carbon (% At) of FS6 sample after adsorption and desorption of methylene blue based on the XPS analysis.

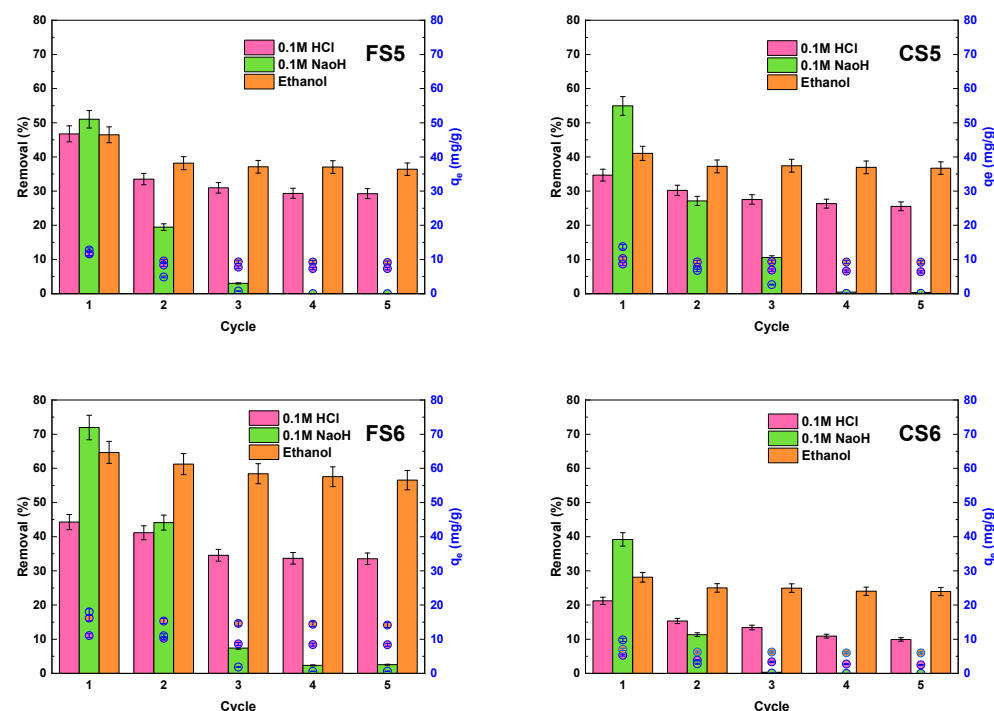


Figure 14. Regeneration performance of bioadsorbent samples after methylene blue adsorption: removal efficiency (bars) and adsorption capacity (dots) (adsorbent mass: 200 mg, volume of dye solution: 0.1 dm³; dye concentration: 50 mg/dm³).

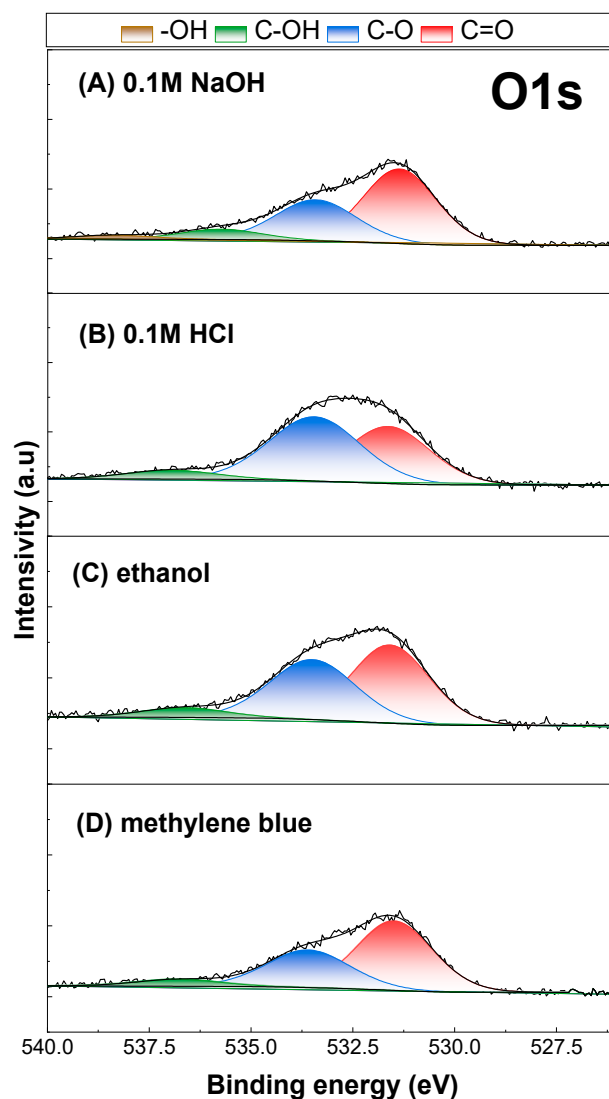


Figure 15. The XPS oxygen (O1s) spectra of FS6 sample after adsorption and desorption of methylene blue.

Table 6. Relative contents of oxygen and carbon (% At) of FS6 sample after adsorption and desorption of methylene blue.

Element	0.1 M NaOH	0.1 M HCl	Ethanol	Methylene Blue
C (% At)	84.63	84.66	84.55	86.00
N (% At)	6.15	4.68	3.82	5.31
O (% At)	9.22	10.66	11.63	8.69

In the first regeneration cycle, NaOH yielded the highest dye desorption efficiency, likely due to its strong alkaline nature, which promotes electrostatic repulsion between the hydroxide ions and the positively charged methylene blue molecules. However, in the next cycles, a significant decline in dye removal efficiency was observed with NaOH. This behavior indicates that the observed effect is not solely related to reversible desorption, but also to chemical alterations occurring on the adsorbent surface upon exposure to a strong base. NaOH has been demonstrated to deprotonate or degrade surface functional groups (e.g., carboxylic or hydroxyl groups) [49]. Furthermore, it has been shown to shift the surface pH and potentially alter the point of zero charge (pH_{pzc}). The consequence of these processes is an excessively negative surface charge. Although this negative charge

initially enhances methylene blue adsorption due to stronger electrostatic attraction, it can also result in site oversaturation or shielding effects, reducing the availability of effective adsorption sites over time.

In contrast, HCl and ethanol exert milder effects on the adsorbent surface; HCl helps maintain active adsorption sites by protonating surface groups, while ethanol facilitates dye removal through physical desorption without significant chemical modification. As a result, the decrease in removal efficiency across cycles is minimal when using HCl and ethanol, indicating better stability and regeneration compatibility. Ethanol proved to be the most effective solvent for desorption due to its unique physicochemical properties, which facilitate the removal of methylene blue molecules from the adsorbent surface without causing significant chemical alterations to the adsorbent itself. As a polar organic solvent with both hydrophilic (-OH group) and hydrophobic (ethyl group) characteristics, ethanol can interact with a wide range of dye molecules through hydrogen bonding, van der Waals forces, and dipole–dipole interactions [50]. This allows ethanol to effectively solubilize methylene blue, disrupting the physical and electrostatic interactions between the dye and the adsorbent surface. Unlike strong acids or bases, ethanol does not modify the surface charge or functional groups of the adsorbent, thereby preserving its structural integrity and adsorption capacity over multiple cycles. Additionally, ethanol's relatively low surface tension and viscosity facilitate deeper penetration into the adsorbent's pores, enhancing desorption efficiency. Its mild nature ensures that no residual ions are left behind, minimizing surface saturation or fouling effects. These combined factors make ethanol a highly effective and non-destructive eluent for regenerating adsorbents in dye removal processes.

An analysis of Figure 15 reveals that the most significant alterations in the O1s spectrum are evident following desorption with a 0.1 M NaOH solution. The peak originating from the C-O bond is significantly smaller than in the other spectra. In addition, a minor peak at approximately 533 (eV) can be discerned, which may indicate the presence of a less electronegative form of oxygen, such as the -OH group. Moreover, a comparison of spectra A and D from sample FS6 with adsorbed dye reveals that they exhibit the greatest similarity to each other. XPS analysis confirmed the presence of nitrogen on the surface of the sheep samples, which was not present in the pure adsorbent. Therefore, it can be interpreted as an indicator of the amount of adsorbed dye. A thorough examination of the data exhibited in Table 6 reveals that desorption facilitated by 0.1 M NaOH results in the detection of the maximum quantity of nitrogen (6.15%) on the surface of the adsorbent. This finding clarifies the apparent contradiction between high initial desorption efficiency and poorer long-term reusability. The elevated nitrogen content indicates incomplete removal of methylene blue or the formation of strongly retained dye residues following alkaline treatment. Simultaneously, NaOH may induce irreversible surface modifications or partial pore obstruction, which progressively reduce the number of accessible active sites. Therefore, the reduced performance in subsequent cycles reflects cumulative surface degradation rather than effective regeneration.

Following a thorough examination of the results obtained after the five desorption cycles, it can be concluded that ethanol was the most effective eluent, achieving the highest removal of the dye after each cycle.

2.4. Mechanism

The adsorption of methylene blue on chemically activated biocarbons derived from fennel and caraway seeds is predominantly governed by physical interactions arising from the adsorbents' surface chemistry and porous structure. XPS, isotherm, kinetic and thermodynamic analyses indicate that MB removal proceeds via multilayer physisorption,

driven by van der Waals forces, electrostatic interactions and diffusion into a heterogeneous pore system. Figure 16 presents a proposed adsorption mechanism. The surface functional groups identified by XPS—primarily C–O, C=O, and C–OH—modulate the electronic environment of the carbon matrix and determine its polarizability. Activation at a higher temperature (600 °C) reduces thermally labile oxygen-containing acidic groups and increases the relative abundance of basic sites [51]. This results in a more electron-rich and polarizable surface. These changes enhance London dispersion interactions between the π -electron system of MB and the graphitic domains of the biochar [52]. The π – π satellite peaks detected in the C1s spectra further support the presence of aromatic domains capable of interacting with the planar aromatic structure of MB through π – π stacking. These non-specific electronic interactions are characteristic of physical adsorption and do not involve the formation of chemical bonds. Electrostatic attraction also plays a role in the uptake of MB, particularly under alkaline conditions. As the pH increases, the deprotonation of surface oxygen groups generates negatively charged sites that attract the cationic MB molecules. This enhances the initial affinity of the dye for the carbon surface, but the process remains fully reversible, consistent with the weak adsorption energies and minimal activation energy implied by the endothermic, low-enthalpy thermodynamic parameters. The positive ΔH° values indicate that MB adsorption benefits from increased molecular mobility and pore accessibility at elevated temperatures, which is typical of physisorption, where external and pore diffusion are rate-limiting processes rather than bond formation.

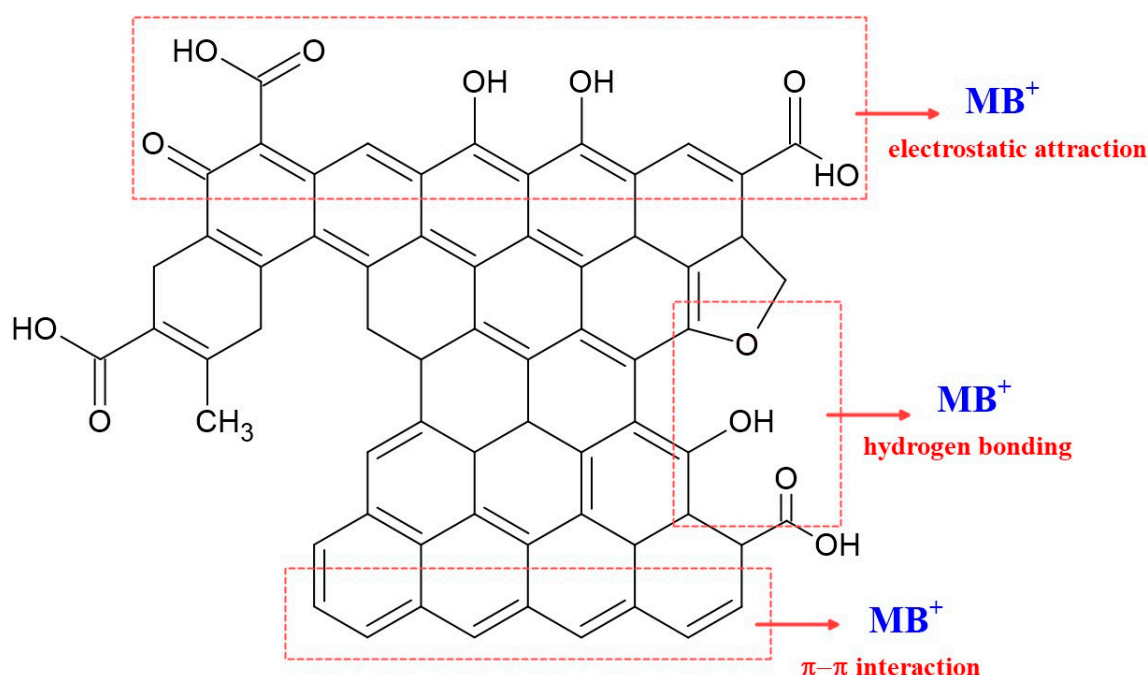


Figure 16. Proposed adsorption mechanism.

The porous texture of the adsorbents plays a central role in providing a physical volume for methylene blue accommodation. Once MB molecules have occupied the primary adsorption sites on the pore walls, subsequent adsorption occurs in additional layers due to MB–MB cohesive interactions. This is reflected in the fit of the Freundlich model. This multilayer behaviour is further reinforced by the relatively high $1/n$ values, which indicate heterogeneous adsorption energies arising from a distribution of pore sizes and surface domains.

Kinetic analysis suggests that the adsorption rate is controlled by the physical transport of methylene blue rather than surface reaction. The suitability of the pseudo-second-order

model indicates that the rate is limited by the availability of active surface sites and the diffusion of methylene blue molecules towards them. The contribution of intraparticle diffusion observed in the multi-linear plots signifies that MB migrates progressively from the external surface into the internal pore network. This is consistent with porous carbon materials, in which diffusion resistance plays a significant role.

Desorption studies also confirm the predominance of physisorption. Ethanol, a mild solvent that disrupts weak van der Waals and dipole-induced dipole interactions without altering the surface chemistry, achieved the most effective regeneration. Conversely, although NaOH was initially efficient, it chemically altered the adsorbent surface rather than reversing adsorption forces, demonstrating that the binding of methylene blue is not chemical in nature.

3. Materials and Methods

3.1. Precursor and Biochar Samples Preparation

The precursors employed in the production of biochar were fennel seeds (FS) and caraway seeds (CS) (Herbapol, Lublin, Poland), which did not meet the quality control standards and were thus considered waste from the herbal industry. The volatile content of caraway seed was found to be 6.69 wt.%, with a moisture content of 5.58 wt.% and an ash content of 4.67 wt.%. For fennel seed, the volatile matter content was 7.9 wt.%, the moisture content was 5.3 wt.%, and the ash content was 5.8 wt.%. Both precursors were subjected to a 24-h drying process at 105 °C, then carbonized at 300 °C for 30 min. in a nitrogen atmosphere with a flow rate of 270 cm³/min. Subsequently, the materials were impregnated with sodium carbonate at a ratio of precursor to activator of 1:2. Thereafter, the carbonized and impregnated materials were subjected to activation at 500 °C (FS5, CS5) and 600 °C (FS6, CS6) for a duration of 15 min. in a nitrogen atmosphere, with a flow rate of 270 cm³/min. Each process was carried out using a microwave oven (Phoenix, CEM Corporation, Matthews, IL, USA) and the temperature rate was 10 °C/min. The resulting biocarbon samples were then washed with hydrochloric acid and subsequently rinsed with boiling distilled water. Following this, the materials were dried until a solid mass was obtained, then sieved through a 0.09 mm sieve and homogenized.

The yield of FS5 biocarbon was 23.03%, CS5 19.47%, FS6 21.30%, and CS6 17.21%. The adsorption properties of the activated carbons were evaluated using an aqueous solution of methylene blue. The remaining chemicals utilised in this study were obtained from Sigma-Aldrich (Burlington, MA, USA), with the assurance of analytical grade.

3.2. Characterization of Resulted Biocarbon Samples

The textural properties of the obtained samples were assessed through nitrogen adsorption/desorption isotherm measurements, following procedures described in previous study [53].

To determine the content of surface oxygen functional groups with basic and acidic properties in the precursor and biochar materials, the Boehm method was employed (Boehm, 1994) [54]. The point of zero charge (pH_{pzc}) of each activated carbon was determined using a batch equilibration method. Six solutions of 0.1 M NaCl were prepared. The pH of these solutions was adjusted stepwise from 2 to 12 using 0.1 M HCl or NaOH. To remove dissolved CO₂ and stabilize the pH, nitrogen gas was bubbled through the solutions. Then, 0.2 g of activated carbon was added to 50 cm³ portions of the NaCl solutions and the suspensions were stirred for 24 h. After equilibration, the final pH (pH_{final}) was measured, and the pH_{pzc} was determined from the intersection point of the curve plotted between the initial pH (pH_{initial}) and pH_{final}.

X-ray photoelectron spectroscopy (XPS) was performed using a Phoibos150 NAP analyser (Specs, Berlin, Germany). The analytical chamber was operated in vacuum at a pressure close to 5×10^{-9} mbar, and the sample was irradiated with nonmonochromatic Al K α radiation (1486.6 eV). Any charge that occurred during the measurements (due to incomplete neutralization of ejected surface electrons) was compensated for by rigidly shifting the entire spectrum by the distance needed to set the C1s binding energy assigned to the carbon to an assumed value of 284.8 eV.

Determination of the iodine number was ascertained in accordance with the ASTM D4607-94 standard. Standard ash analysis was performed in accordance with the ASTM D2866-94 (2004) method.

Scanning Electron Microscope (SEM) images were acquired with a working distance of 7.7 mm and an accelerating voltage of 25.0 kV, with digital image capture managed through the DISS system.

X-ray diffraction patterns were recorded at room temperature with a step size of 0.02° for the low-angle range and 0.05° for the high-angle range. The XRD diffractograms were analysed using Kalvados software.

3.3. Adsorption Studies

Methylene blue was selected as the model dye for this study. Due to its photosensitivity, all adsorption experiments were conducted in the dark. In a typical experiment, 20 mg of each biochar sample was added to 0.05 dm^3 of an aqueous methylene blue solution with initial concentrations ranging from 5 to 30 mg/dm^3 . The suspensions were agitated at 300 rpm using a Heidolph laboratory shaker at room temperature ($22 \pm 1^\circ \text{C}$) for 24 h. After adsorption, the residual dye concentration was determined by measuring the absorbance at 665 nm with a dual-beam UV–Vis spectrophotometer (Cary Bio 100, Varian). A calibration curve was constructed in the concentration range of $0\text{--}10 \text{ mg/dm}^3$ (Figure 17), which showed a linear relationship between absorbance and concentration. Samples exhibiting absorbance values exceeding the linear range were appropriately diluted tenfold prior to measurement to ensure accurate quantification. For all subsequent experiments, the adsorbent mass, solution volume, initial dye concentration, and agitation speed were fixed at 20 mg, 0.05 dm^3 , 20 mg/dm^3 , and 300 rpm, respectively. The solution pH was not adjusted unless otherwise stated.

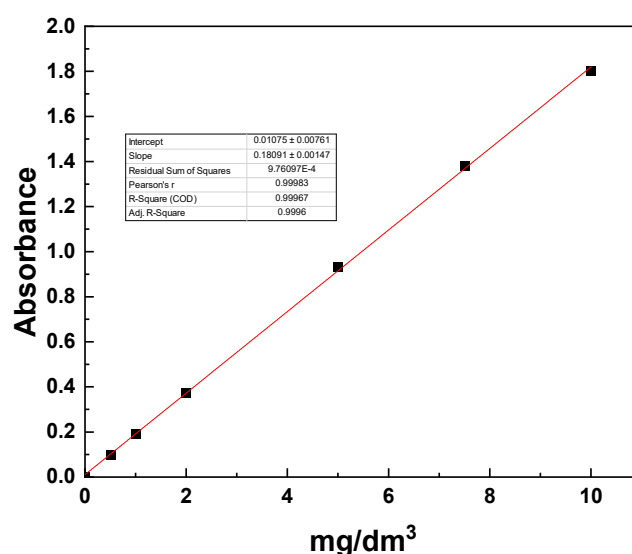


Figure 17. The calibration curve for methylene blue.

Table 7 exhibits applied equations and corresponding parameters. The adsorption capacities (q_e , mg/g) of the biochar samples were calculated using the Formula (1) shown in Table 6. The linear forms of the Langmuir and Freundlich equations were employed to determine a suitable model for the adsorption of dye on biochar. The Langmuir isotherm can be represented by the linear Equation (2). The Freundlich isotherm can be represented by the linear Equation (3). Thermodynamic parameters were determined using Formulas (4)–(6). In order to characterize the kinetics of methylene blue adsorption on the biocarbon samples, four models were employed: the pseudo-first-order model (7), the pseudo-second-order (8), the Elovich model (9) and intraparticle diffusion (10).

Table 7. Applied equations and corresponding parameters.

Equation		Parameter
(1)	$q_e = \frac{(C_0 - C_e)}{m} \times V$	C_0 —the initial concentration (mg/dm ³) of the dye in solution; C_e —the equilibrium concentration (mg/dm ³) of the dye in solution; m —the mass of the biocarbon sample (g); V —the volume of the solution (dm ³);
(2)	$\frac{1}{q_e} = \frac{1}{q_{max}} + \frac{1}{K_L q_{max}} \times \frac{1}{C_e}$	q_e —the equilibrium amount of adsorbed dye (mg/g); K_L —Langmuir equilibrium constant (dm ³ /mg); q_{max} —the maximum adsorption capacity of the adsorbent (mg/g);
(3)	$\log q_e = \log K_F + \frac{1}{n} \log C_e$	K_F —Freundlich equilibrium constant (mg/g(dm ³ /mg) ^{1/n}); $1/n$ —the adsorption intensity constant;
(4)	$\Delta G^\circ = -RT \ln K_d$	ΔG° —Gibbs free energy; R —universal constant (8.314 J/mol × K);
(5)	$\Delta G^\circ = \Delta H^\circ - T \Delta S^\circ$	T —temperature (K); ΔH° —enthalpy change;
(6)	$\ln K_d = \frac{\Delta S^\circ}{R} - \frac{\Delta H^\circ}{RT}$	ΔS° —entropy change; K_d —thermodynamic equilibrium constant;
(7)	$\log(q_e - q_t) = \log q_e - \frac{k_1}{2.303} t$	q_e —the equilibrium amount of adsorbed dye (mg/g); q_t —the amount of adsorbed dye over time (mg/g);
(8)	$\frac{t}{q_t} = \frac{1}{k_2 q_e} + \frac{t}{q_e}$	t —the process time (min) values in the referenced study; k_1 —the pseudo-first-order adsorption constant (1/min); k_2 —the pseudo-second-order adsorption constant (g/mg × min);
(9)	$q_t = \frac{1}{\beta} \ln(1 + \alpha \beta t)$	α —the Elovich initial sorption rate constant (mg/g × min); β —the Elovich desorption constant (g/mg);
(10)	$q_t = k_{id} t^{1/2} + C$	k_{id} —the intraparticle diffusion constant (mg/g × min ^{1/2}); C —the intraparticle diffusion model's boundary layer constant (mg/g).

Furthermore, the study investigated the impact of process temperature on the adsorption of an aqueous solution containing methylene blue. In order to ascertain the effect of the temperature of the aqueous dye solution on the adsorption efficiency, a series of three samples (20 mg) of each carbon were prepared and then immersed analogously to the weighing test. The samples were then subjected to a 24-h shaking period at temperatures of 25, 35, 45, 55 and 65 °C. Subsequently, the samples were subjected to centrifugation in a laboratory centrifuge, and spectrophotometric measurements were conducted. Moreover, the study investigated the effect of the pH value (ranging from 4 to 10) of the initial aqueous methylene blue solution on the sorption capacity of the resulting activated carbons.

The effect of ionic strength on methylene blue adsorption was investigated using sodium chloride (NaCl) as the electrolyte. Different ionic strengths were achieved by adding appropriate amounts of NaCl to the MB solutions to obtain final concentrations of 0, 0.01, 0.05, and 0.1 mol/dm³. The dye solutions were prepared first, and the required quantity of NaCl was subsequently added and dissolved completely before the adsorption experiments were initiated.

A desorption study was conducted as a separate set of experiments using three eluents: 0.1 M HCl, NaOH, and ethanol. For this purpose, 0.2 g of each carbon sample was directly weighed and first subjected to a dedicated adsorption step using 100 cm³ of methylene blue solution with an initial concentration of 50 mg/dm³ to ensure saturation of the adsorbent. After reaching adsorption equilibrium, the saturated samples were transferred to a solid-phase extraction (SPE) column and treated with 100 cm³ of the selected eluent. Subsequently, the samples were rinsed with 100 cm³ of distilled water to neutralize the surface, dried, and reused for further adsorption–desorption cycles. The entire procedure was repeated five times. The post-desorption solutions were collected and neutralized prior to disposal in accordance with laboratory safety and waste management procedures. The desorbed solutions were not reused further and were treated as chemical waste.

4. Conclusions

Biocarbon adsorbents derived from fennel and caraway seeds were successfully produced through chemical activation with sodium carbonate using a microwave-assisted method. The process parameters included carbonization at 300 °C and activation at 500 °C and 600 °C. The physicochemical characterisation of samples revealed that the temperature of activation exerts an insignificant influence on the specific surface area and sorption properties. The specific surface area of the materials obtained from caraway seed was between 109 and 146 mg/g and for fennel seed, 117–154 m²/g. The most effective sorption properties were demonstrated by activated carbon obtained from fennel seed at a temperature of 600 °C. The material exhibited a maximum sorption capacity for methylene blue of 32 mg/g. The analysis of the nature of adsorption using the Langmuir and Freundlich models revealed that the methylene blue adsorption data exhibited a superior fit for the Freundlich model. This phenomenon is indicative of the formation of a multilayer of methylene blue molecules on the activated carbon surfaces during the adsorption process. The results showed that the sorption capacity of the tested biocarbon materials increased with rising pH in the case of methylene blue. The adsorption kinetics are most accurately described by the pseudo-second-order model. The results of the thermodynamic study indicate that the methylene blue adsorption process is physical in nature, as evidenced by the low values of ΔH° . Although the Na₂CO₃-activated biochars exhibit moderate adsorption capacities compared to strongly activated carbons, they demonstrate functional dye removal performance achieved under lower-temperature microwave conditions, supporting their potential as energy-efficient and environmentally sustainable adsorbents.

Future research should focus on investigating the use of more potent agents, such as K₂CO₃ or H₃PO₄, which may yield materials with enhanced adsorption performance. Additionally, optimizing other process parameters, such as impregnation ratio, heating rate, and residence time, may further enhance the structural and functional properties of the resulting adsorbents. Expanding the scope of contaminants studied beyond methylene blue could also help assess the versatility and practical application of these biocarbon materials in water treatment technologies.

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